

Software Requirements Specification (SRS)

for

Smart Warehouse Automation System (SWAS)

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Software Requirements Specification

Table of Contents

- 1. Introduction
 - 1.1 Purpose
 - 1.2 Document Conventions
 - 1.3 Intended Audience and Reading Suggestions
 - 1.4 Product Scope
 - 1.5 References
- 2. Overall Description
 - 2.1 Product Perspective
 - 2.2 User Characteristics
 - 2.3 Operating Environment
 - 2.4 Design and Implementation Constraints
 - 2.5 Assumptions and Dependencies
- 3. Specific Requirements
 - 3.1 External Interface Requirements
 - 3.2 Functional Requirements (By Module)
 - 3.3 Performance Requirements
 - 3.4 Logical Database Requirements
 - 3.5 Software System Attributes (Non-Functional)
- 4. Appendixes
 - A. Glossary of Terms

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the complete functional and non-functional requirements for the Smart Warehouse Automation System (SWAS). This document will serve as the primary reference for the design, development, quality assurance, and deployment teams. It ensures all stakeholders have a unified understanding of the system's capabilities before development commences.

1.2 Document Conventions

- **Must:** A mandatory requirement that must be implemented.
- **Should:** A desirable requirement that may be deferred if time/resources are limited.
- **Shall:** Indicates a binding contractual requirement.
- **R#:** Unique requirement identifier (e.g., R-101, R-202) for traceability.

1.3 Intended Audience and Reading Suggestions

- **Development Team:** Must read Sections 3 (Specific Requirements) and 4 (Appendixes) in detail.
- **Project Managers:** Focus on Sections 1 (Introduction) and 2 (Overall Description) for scope and planning.
- **Quality Assurance (QA) Team:** Use Section 3.2 (Functional Requirements) to create test cases.
- **System Architects:** Concentrate on Section 3.1 (Interfaces) and 3.3 (Performance).

1.4 Product Scope

The SWAS is a software platform designed to manage and control a fully automated warehouse. It will integrate with:

- Autonomous Mobile Robots (AMRs) for transporting goods.
- Conveyor belt systems for sorting.
- Automated Storage and Retrieval Systems (AS/RS) for racking pallets.
- Barcode/RFID scanners for inventory tracking.

The system replaces a legacy manual-pick system, aiming to increase throughput from 500 orders/hour to 2,500 orders/hour while reducing human error in inventory counting to < 0.5%.

1.5 References

- ISO/IEC/IEEE 12207: Systems and software engineering – Software life cycle processes.
- IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications.
- Internal Warehouse Operations Manual (WOM v3.2).

2. Overall Description

2.1 Product Perspective

External System	Interface Type	Data Exchanged
Enterprise Resource Planning (ERP) - SAP	REST API (HTTPS)	Incoming customer orders, inventory valuation.
Fleet Management System (FMS)	MQTT Protocol (IoT)	Robot health status, battery levels, assigned tasks.
HR/Payroll System	SFTP (File Transfer)	Employee shift schedules and performance metrics.

2.2 User Characteristics

User Role	Technical Level	Responsibilities
Warehouse Manager	Medium	Monitors real-time dashboards, generates productivity reports, manages exception handling.

User Role	Technical Level	Responsibilities
System Administrator	High	Configures robot paths, updates firmware, manages user permissions, and performs backups.
Inventory Controller	Medium	Manages stock counts, reconciles physical vs. system inventory, handles damaged goods.
Maintenance Technician	Low	Views error logs and robot status; receives maintenance alerts.

2.3 Operating Environment

- Servers: On-premise dedicated servers (failover cluster) running Linux (Ubuntu 22.04 LTS).
- Network: High-speed local Gigabit Ethernet with VLAN segmentation for operational and administrative traffic.
- Database: Microsoft SQL Server 2022 (relational) and InfluxDB (time-series data for robot telemetry).
- Client Workstations: Windows 11 PCs with Chrome browsers for the Web UI.

2.4 Design and Implementation Constraints

- C-01: All software shall be developed in Java 17 (Spring Boot) for backend microservices.
- C-02: The frontend interface shall be a Single Page Application (SPA) built with React.js.
- C-03: The system must be capable of operating in a Degraded Mode (50% capacity) if the primary server fails.
- C-04: No open-source AI/ML libraries shall be used for pathfinding algorithms without prior security audit.

2.5 Assumptions and Dependencies

Assumptions:

A-01: Warehouse floor Wi-Fi coverage is 100% uninterrupted (for robot communication).

A-02: Physical barcodes on all inventory items are clearly legible and standardized.

A-03: Power supply to the servers has an Uninterruptible Power Supply (UPS) backup.

Dependencies:

D-01: The system depends on the third-party Mobotix robot SDK for hardware commands.

D-02: Integration with SAP ERP is contingent on SAP providing an updated API key.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 Hardware Interfaces

- HI-01: The server must interface with barcode scanners (USB connected) to accept inventory data.
- HI-02: The system should support integration with thermal label printers (Zebra) for generating shipping labels.

3.1.2 Software Interfaces

- SI-01 (Robot Interface): The SWAS must send navigation waypoints (X,Y coordinates) to the FMS via MQTT. The payload shall be JSON format: `{"robot_id": "R-12", "dest_x": 45.5, "dest_y": 12.3}`.
- SI-02 (ERP Interface): The SWAS must consume incoming orders from SAP via a RESTful GET endpoint called `/api/orders/pending`. Response time must be under 500ms.

3.1.3 User Interfaces (UI)

- UI-01: The main dashboard must display a live 3D heatmap of robot activity and zone congestion.
- UI-02: The Inventory Screen must provide a search filter by SKU, Batch Number, or Expiry Date.

3.2 Functional Requirements (By Module)

Module 1: Order Receiving & Processing

- R-101: The system shall automatically ingest new customer orders from the ERP every 5 minutes via the scheduled polling mechanism.
- R-102: The system must validate that the quantity requested for each SKU is available in inventory before accepting the order.
- R-103: Upon validation, the system shall group orders into "waves" (clusters) based on proximity of items to optimize robot travel distance.

Module 2: Robot Task Management

- R-201: The system shall assign specific pick/drop tasks to the nearest available AMR robot based on real-time location (GPS coordinate data).
- R-202: The system must prioritize "Express" orders (Priority Level 1) over standard orders.
- R-203: If a robot fails to complete a task within 10 minutes, the system shall retask the job to another robot and mark the original robot as "Out of Service" for diagnostics.

Module 3: Inventory Management

- R-301: The system must perform a perpetual inventory count, updating stock levels in real-time every time a robot picks or drops an item.
- R-302: The system must generate a "Cycle Count" report at the end of every shift, highlighting discrepancies between system stock and physical scans.
- R-303: The system shall block the processing of orders for any SKU with stock < reorder threshold (minimum level) and send an alert to the Inventory Controller.

Module 4: Reporting & Analytics

- R-401: The system must generate a daily productivity report showing orders fulfilled per hour, average robot travel time, and downtime.
- R-402: The system must maintain an audit log of all user actions (Login, Order Cancellation, Inventory Adjustments) for compliance tracing.

Module 5: Alerts & Notifications

- R-501: The system must send push notifications to the Maintenance Technician's dashboard if a robot's battery falls below 15%.
- R-502: The system must trigger a "System Alert" (Visual Red Banner) if the conveyor belt buffer exceeds 90% capacity.

3.3 Performance Requirements

- PR-01: The system must process an incoming order from ERP and generate a robot task in under 2 seconds.
- PR-02: The real-time inventory database must support up to 1,500 transactions per second (TPS) during peak hours.
- PR-03: The 3D dashboard UI must refresh robot positions with a latency of less than 500ms (near real-time).
- PR-04: The System must support a minimum of 50 concurrent robot control sessions without degradation.

3.4 Logical Database Requirements

- DB-01: The system shall maintain the following core relational tables:
 - Robot_Table (Robot_ID, Model, Battery_Level, Last_Location, Status)
 - Inventory_Table (SKU, Description, Location_Bin, Quantity, Reorder_Level)
 - Order_Table (Order_ID, Customer, SKU_List, Priority, Timestamp)
- DB-02: All time-stamped telemetry data (robot speed, temperature, payload weight) shall be stored in a separate InfluxDB time-series database with a retention policy of 30 days.

3.5 Software System Attributes (Non-Functional)

3.5.1 Security

- SEC-01: User access must be governed by Role-Based Access Control (RBAC). Default role is "Read-Only".
- SEC-02: All API communication between SWAS and external systems must be encrypted using TLS 1.3.
- SEC-03: User passwords must be stored using PBKDF2 hashing with a minimum of 10,000 iterations.

3.5.2 Reliability

- REL-01: The system must have an uptime of 99.95% (excluding scheduled maintenance windows on Sundays 2:00 AM - 4:00 AM).
- REL-02: In the event of a database crash, the system must failover to the secondary replica within 60 seconds.

3.5.3 Usability

- US-01: The Web UI must be accessible to color-blind users (avoid red/green only alerts; use icons/shapes).
- US-02: All critical action buttons (e.g., "Cancel Order," "Shutdown Robot") must require a confirmation dialog to prevent accidental clicks.

3.5.4 Maintainability

- MAINT-01: The system must maintain structured log files (ERROR, WARN, INFO) with a standard JSON format for integration with ELK stack.
- MAINT-02: The codebase must maintain a minimum test coverage of 80% for unit tests and 90% for critical path components.

4. Appendixes

Appendix A: Glossary of Terms

Term	Definition
AMR	Autonomous Mobile Robot: a robot that moves independently without fixed paths.
AS/RS	Automated Storage and Retrieval System: robotic cranes that put pallets on high racks.
Cycle Count	A periodic inventory count of a specific zone rather than the entire warehouse.
FMS	Fleet Management System: the software that directly drives the robot hardware.
SKU	Stock Keeping Unit: a unique identifier for each distinct product and service.
Wave Picking	The process of grouping multiple orders together to pick items for several orders simultaneously.

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